Let $\mathrm{row}(X)$ denote the number of rows of a matrix X. Let

$$A \in \mathbb{R}^{m\times d} \tag{1}$$

be the full data matrix, let A_w denote the current row-window, and let $B \in \mathbb{R}^{r\times d}$ denote the sketch carried from the previous windows. Write

$$A_w \in \mathbb{R}^{k\times d}, \qquad k = \mathrm{row}(A_w), \qquad k > 1, \tag{2}$$

$$t_w := (\#\text{ rows already seen up to and including the current window}), \qquad v \in \mathbb{S}^{d-1}, \tag{3}$$

and write

$$h_2\left(\frac{Xv}{\|Xv\|_2}\right) = -\log\left(\frac{\|Xv\|_4^4}{\|Xv\|_2^4}\right) \tag{4}$$

for any matrix X with a nonzero response on v. We use the score

$$\mathrm{score}_w(v) := \|M_w v\|_2 \left(\frac{t_w}{d}\right)^{-\frac{\widehat{h}_{2,w}(v)}{4\log t_w}}, \tag{5}$$

where

$$M_w := \begin{pmatrix} B \\ A_w \end{pmatrix}, \tag{6}$$

and, when

$$B = SV^\top, \qquad V = [v_1, \dots, v_r], \qquad S = \mathrm{diag}(s_1, \dots, s_r), \tag{7}$$

with carried entropy values $h_1, \dots, h_r$, the streamed entropy surrogate is

$$e_w(v) := \sum_{j=1}^{r} s_j^2 (v_j^\top v)^2 + \|A_w v\|_2^2, \tag{8}$$

$$q_w(v) := \sum_{j=1}^{r} s_j^4 e^{-h_j} (v_j^\top v)^4 + \|A_w v\|_4^4, \tag{9}$$

$$\widehat{h}_{2,w}(v) := -\log \frac{q_w(v)}{e_w(v)^2}. \tag{10}$$

For the first window, where $B = 0$ and $M_1 = A_1$, this reduces to the exact entropy

$$\widehat{h}_{2,1}(v) = h_2\left(\frac{A_1 v}{\|A_1 v\|_2}\right). \tag{11}$$

Algorithm overview. The method processes the matrix one row-window at a time.

For the first window, there is no previous sketch, so we set

$$B = 0, \qquad M_1 = A_1, \tag{12}$$

and define the first-window objective

$$\mathrm{score}_1(v) := \|A_1 v\|_2 \left(\frac{\mathrm{row}(A_1)}{d}\right)^{-\frac{h_2\left(\frac{A_1 v}{\|A_1 v\|_2}\right)}{4 \log \mathrm{row}(A_1)}}. \tag{13}$$

The retained directions are computed sequentially by solving

$$v_j \in \operatorname*{arg\,max}_{v \in \mathbb{S}^{d-1},\ v_i^\top v = 0\ \forall i<j} \mathrm{score}_1(v). \tag{14}$$

After the selected directions are accepted, the sketch carried to the next window is simply

(15)

$$h_j^{(1)} := \widehat{h}_{2,1}(v_j), \qquad s_j^{(1)} := \|M_1 v_j\|_2 \ \ (j = 1, \dots, r), \qquad h^{(1)} = (h_1^{(1)}, \dots, h_r^{(1)}), \qquad S_1 = \mathrm{diag}(s_1^{(1)}, \dots, s_r^{(1)}),$$

(16)

so that each accepted direction v_j is carried together with its gain $s_j^{(1)}$ and entropy statistic $h_j^{(1)}$.

For a later window w, the method receives the previous sketch

$$V_{w-1} = [v_1, \dots,$$

(17)

$$S_{w-1} = \mathrm{diag}(s_1^{(w-1)}, \dots, s_r^{(w-1)}), \qquad h^{(w-1)} = (h_1^{(w-1)}, \dots, h_r^{(w-1)}), \qquad B = B^{(w-1)} = S_{w-1} V_{w-1}^\top$$

(18)

where

$$s_j^{(w-1)} := \|M_{w-1} v_j\|_2, \qquad h_j^{(w-1)} := \widehat{h}_{2,w-1}(v_j), \qquad j = 1, \dots, r. \tag{19}$$

The streamed operator M_w, the surrogate moments $e_w(v), q_w(v)$, and the objective $\widehat{\mathrm{score}}_w(v)$ are the quantities defined above from this carried sketch and the current block. The retained directions are computed sequentially from

$$v_j^{(w)} \in \operatorname*{arg\,max}_{v \in \mathbb{S}^{d-1},\ (v_i^{(w)})^\top v = 0\ \forall i<j} \widehat{\mathrm{score}}_w(v). \tag{20}$$

After solving for the updated directions, the next sketch is again the direct singular-factor summary

$$V_w = [v_1^{(w)}, \dots, v_r^{(w)}] \qquad s_j^{(w)} := \|M_w v_j^{(w)}\|_2 \ \ (j = 1, \dots, r), \qquad h_j^{(w)} := \widehat{h}_{2,w}\left(v_j^{(w)}\right), \qquad S_w = \mathrm{diag}(s_1^{(w)}, \dots,$$

(21)

Thus a middle window sees a compact summary of all previous windows through $S_w V_w^\top$, rather than raw past rows.

Derivation of the score. Let

$$z = Xv \tag{22}$$

be a row-restricted response and let

$$x = Av \tag{23}$$

be the response of the full matrix. We use the normalized-entropy heuristic

$$\frac{h_2\left(\frac{x}{\|x\|_2}\right)}{\log d} \approx \frac{h_2\left(\frac{z}{\|z\|_2}\right)}{\log \mathrm{row}(X)}, \tag{24}$$

that is,

$$h_2\left(\frac{x}{\|x\|_2}\right) \approx h_2\left(\frac{z}{\|z\|_2}\right) \frac{\log d}{\log \mathrm{row}(X)}. \tag{25}$$

Using

$$h_2\left(\frac{y}{\|y\|_2}\right) = 4 \log \|y\|_2 - 4 \log \|y\|_4, \tag{26}$$

and $\|z\|_4 \leq \|x\|_4$ for a restriction z of x, we obtain

$$4 \log \frac{\|z\|_2}{\|x\|_2} \leq h_2\left(\frac{z}{\|z\|_2}\right) - h_2\left(\frac{x}{\|x\|_2}\right). \tag{27}$$

Substituting the heuristic entropy scaling gives

$$\log \frac{\|z\|_2}{\|x\|_2} \lesssim -\frac{h_2\left(\frac{z}{\|z\|_2}\right)}{4} \frac{\log(d/\mathrm{row}(X))}{\log \mathrm{row}(X)}, \tag{28}$$

hence

$$\frac{\|Xv\|_2}{\|Av\|_2} \lesssim \left(\frac{\mathrm{row}(X)}{d}\right)^{\frac{h_2\left(\frac{Xv}{\|Xv\|_2}\right)}{4 \log \mathrm{row}(X)}}. \tag{29}$$

This motivates the reciprocal entropy factor

$$\left(\frac{\mathrm{row}(X)}{d}\right)^{-\frac{h_2\left(\frac{Xv}{\|Xv\|_2}\right)}{4 \log \mathrm{row}(X)}}. \tag{30}$$

Thus, for an exact window matrix X, we use

$$\mathrm{score}_X(v) := \|Xv\|_2 \left(\frac{\mathrm{row}(X)}{d}\right)^{-\frac{h_2\left(\frac{Xv}{\|Xv\|_2}\right)}{4 \log \mathrm{row}(X)}}. \tag{31}$$

First-window special case. If there is no previous sketch, then $B = 0$ and $M_1 = A_1$, so the first-window score is the exact objective already introduced above. Using

$$h_2\left(\frac{A_1 v}{\|A_1 v\|_2}\right) = 4\log\|A_1 v\|_2 - 4\log\|A_1 v\|_4, \tag{32}$$

this becomes

$$\mathrm{score}_1(v) = \|A_1 v\|_2^{1-c_1}\|A_1 v\|_4^{c_1}, \qquad c_1 := \frac{\log(k/d)}{\log k}. \tag{33}$$

Inductive streamed form. For later windows $w \geq 2$, use the carried sketch and entropy statistics already introduced above. Writing $a = V^\top v$ for the coefficients of v in the stored basis, the previously defined moment surrogates become

$$e_w(v) = \sum_{j=1}^{r} a_j^2 s_j^2 + \|A_w v\|_2^2, \qquad q_w(v) = \sum_{j=1}^{r} a_j^4 s_j^4 e^{-h_j} + \|A_w v\|_4^4. \tag{34}$$

Since $B = SV^\top$ and $M_w = \begin{pmatrix} B \\ A_w \end{pmatrix}$ by the earlier definitions, we have

$$\|M_w v\|_2^2 = \|Bv\|_2^2 + \|A_w v\|_2^2 = e_w(v). \tag{35}$$

Hence the inductive score is exactly the previously defined $\widehat{\mathrm{score}}_w(v)$.

Simplified log form. Let

$$\gamma_w := \frac{\log(t_w/d)}{2\log t_w}. \tag{36}$$

Then

$$\log\widehat{\mathrm{score}}_w(v) = \frac{1}{2}\log\|M_w v\|_2^2 + \gamma_w \log q_w(v) - 2\gamma_w \log e_w(v). \tag{37}$$

Since $\|M_w v\|_2^2 = e_w(v)$, this can be written equivalently as

$$f_w(v) := \log\widehat{\mathrm{score}}_w(v) = \frac{1}{2}\log e_w(v) + \gamma_w \log q_w(v) - 2\gamma_w \log e_w(v) \tag{38}$$

or

$$f_w(v) = \frac{1-c_w}{2}\log e_w(v) + \frac{c_w}{4}\log q_w(v), \qquad c_w := \frac{2\log(t_w/d)}{\log t_w}. \tag{39}$$

This is exactly the same interpolation between quadratic energy and fourth-moment concentration as in the first window, but now with the old moments represented by the carried sketch.

Exact gradient. For the first window,

$$\nabla f_1(v) = (1 - c_1)\,\frac{A_1^\top A_1 v}{\|A_1 v\|_2^2} + c_1 A_1^\top \left(\frac{(A_1 v)^{\odot 3}}{\|A_1 v\|_4^4}\right). \tag{40}$$

For later windows, differentiate the streamed quantities:

$$\nabla e_w(v) = 2V\big((s^{\odot 2}) \odot a\big) + 2A_w^\top A_w v, \tag{41}$$

$$\nabla q_w(v) = 4V\big((s^{\odot 4} \odot e^{-h}) \odot a^{\odot 3}\big) + 4A_w^\top\big((A_w v)^{\odot 3}\big). \tag{42}$$

Therefore

$$\nabla f_w(v) = \frac{1}{2}\,\frac{\nabla \|M_w v\|_2^2}{\|M_w v\|_2^2} + \gamma_w \frac{\nabla q_w(v)}{q_w(v)} - 2\gamma_w \frac{\nabla e_w(v)}{e_w(v)}. \tag{43}$$

Using $\nabla \|M_w v\|_2^2 = 2M_w^\top M_w v$, this is

$$\nabla f_w(v) = \frac{M_w^\top M_w v}{\|M_w v\|_2^2} + \gamma_w \frac{\nabla q_w(v)}{q_w(v)} - 2\gamma_w \frac{\nabla e_w(v)}{e_w(v)}. \tag{44}$$

This is the full streamed EntropyScore gradient used by the implementation.

Stored-basis form. If

$$q_j := s_j^4 e^{-h_j}, \tag{45}$$

then

$$q_w(v) = \sum_{j=1}^{r} q_j (v_j^\top v)^4 + \|A_w v\|_4^4. \tag{46}$$

So the carried state needed by the next window is naturally

$$(V, s, h) \qquad \text{or equivalently} \qquad (V, s^2, q), \tag{47}$$

with

$$q = s^{\odot 4} \odot e^{-h}. \tag{48}$$

This is exactly the low-rank moment summary used in the streamed objective.

Optimization problem. To extract a direction, solve

$$\max_{v \in \mathbb{S}^{d-1},\ Q^\top v = 0} \widehat{\mathrm{score}}_w(v), \tag{49}$$

where the columns of Q are the previously accepted directions.

Projected full-gradient ascent. Use the exact gradient of the streamed score and project it onto the feasible tangent space:

$$g_{\tan}(v) := \left(I - QQ^\top - vv^\top\right) \nabla f_w(v). \tag{50}$$

Then take

$$\widetilde{v}^+ = v + \eta\, g_{\tan}(v), \qquad v^+ = \frac{(I - QQ^\top)\widetilde{v}^+}{\|(I - QQ^\top)\widetilde{v}^+\|_2}, \tag{51}$$

with $\eta > 0$ chosen by backtracking line search on the true streamed objective.

Basis-expansion interpretation. The expansion variant does not optimize only inside one fixed reduced space. It begins with a seed right subspace and checks whether the full gradient is sufficiently represented there.

If V_{basis} is the current reduced basis and g_{full} is the full-space gradient at the current candidate, the residual direction is

$$r := g_{\text{full}} - V_{\text{basis}} V_{\text{basis}}^\top g_{\text{full}}. \tag{52}$$

When the ratio

$$\frac{\|r\|_2}{\|g_{\text{full}}\|_2} \tag{53}$$

is too large, the basis is enlarged by this missing component and, optionally, by additional Krylov directions generated from

$$(M_w^\top M_w)^t r. \tag{54}$$

Thus the basis is expanded exactly when the current reduced space fails to capture the true score gradient well enough.

Why this is the sketch carried into the next window. After the window solve, the method carries forward the singular-factor sketch

$$B_{\text{next}} := SV^\top, \tag{55}$$

where V collects the accepted right directions and $S = \operatorname{diag}(s_1, \ldots, s_r)$ with $s_j = \|Mv_j\|_2$ for the corresponding carried operator M.

This is the justified object to pass forward for several reasons.

First, the next-window objective needs the old information through the quadratic term

$$\|B_{\text{next}} v\|_2^2 = \|SV^\top v\|_2^2 = \sum_{j=1}^{r} s_j^2 (v_j^\top v)^2, \tag{56}$$

which is exactly the carried second-moment contribution appearing in $e_w(v)$.

Second, together with the stored entropy statistics h_j, the same sketch determines the old fourth-moment term

$$\sum_{j=1}^{r} s_j^4 e^{-h_j} (v_j^\top v)^4 = \sum_{j=1}^{r} q_j (v_j^\top v)^4, \tag{57}$$

which is the carried part of $q_w(v)$.

Third, unlike the forget variant, the expansion method is intentionally modeling both old quadratic energy and old fourth-moment concentration. Therefore the natural next-window summary is the direct spectral sketch together with its per-direction entropy statistic, not a left-projected operator replacement.

Fourth, this carry is consistent with the streamed operator

$$M_{w+1} = \begin{pmatrix} B_{\text{next}} \\ A_{w+1} \end{pmatrix}, \tag{58}$$

used by the next window. The old block enters exactly through the compact low-rank operator $SV^\top$, while the new block contributes its fresh exact moments.

Therefore the correct sketch to pass forward for `entropyscore_expansion` is the direct singular-factor summary

$$B_{\text{next}} = SV^\top, \tag{59}$$

along with the entropy carry variables $h = (h_1, \ldots, h_r)$ and $q = s^{\odot 4} \odot e^{-h}$.